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May 16, 2014

The Quantum PlayStation

How the PS3 is helping physicists develop a theory of quantum gravity

by Graeme Stemp-Morlock

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July 18, 2010



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Like many physicists, Gaurav Khanna is a PlayStation fan. Unlike other physicists, however, he is using the games console in his efforts to answer one of the biggest questions in cosmology: What came before the big bang?

When students learn the standard story about the birth of our universe, they are usually told not to ask what came *before* it. Time, the textbooks say, was created along with space during the big bang. But some physicists think that things aren't quite that straightforward. By throwing out some of the basic tenets of classical physics, they have come up with a picture in which our universe was preceded by another, which crunched down and then bounced outwards again. And according to Khanna, the SONY PlayStation may be the ideal—if seemingly unlikely—tool for peering back into this time before time.

Khanna, at the University of Massachusetts, Dartmouth, is using a network of PS3 consoles and a [\\$15,000 grant from the Foundational Questions Institute](#) to model this bouncing universe. He has been inspired by attempts to unite two of the most successful theories in physics: quantum mechanics, which rules the subatomic world, and general relativity, which describes situations where gravity reigns. So far, physicists have been frustrated in their attempts to find a theory that meshes the two together. Such a theory is needed if we want to understand what happens when immense gravitational forces are concentrated into microscopic volumes of spacetime, such as in black holes or at the birth of our universe.

One candidate for finding the answer is [loop quantum cosmology](#). This theory rejects the assumption of general relativity that spacetime is made up of a continuous fabric; instead it says that spacetime is built up of discrete blocks, rather like a digital image. You may see a smooth picture, but when you zoom in you reveal individual pixels. Luckily, a pixelated universe is easy to simulate. "Computers can handle discrete objects and manipulate them very effectively and very fast," says Khanna. "After all that's their forte."

In this model, our universe did not begin in an infinitely small big bang singularity, because it is impossible to squash spacetime down below some small, but finite value. Instead, the universe bounced out from a tiny volume, which in turn, was the remnant of a prior crunched-down cosmos.

But why use the PS3 to model the origins of this universe? One advantage of the PS3 is that at the time that Khanna began the project it was an open platform, which could run Linux, making scientific programming relatively easy. By hooking together a number of consoles—each with the power to accommodate fast-paced action games—Khanna has created a supercomputer equivalent to nearly 200 normal desktop computers. Not only is the PS3 supercomputer 10 times more cost effective, but it is also 10 times more power efficient, using far less electricity. In fact, SONY were so impressed by Khanna's enterprising idea that some of their R&D division even helped Khanna get the project up and running.



GAURAV KHANNA

University of Massachusetts at Dartmouth

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Gaming Supercomputer

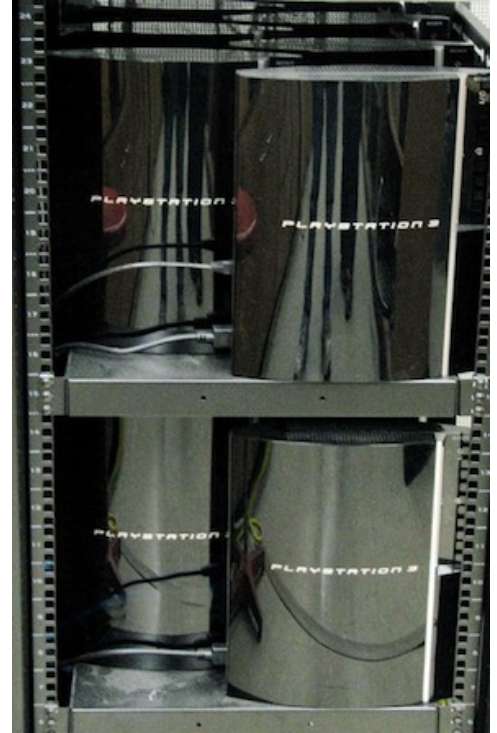
Khanna's gaming supercomputer has already proved a success at simulating the behavior of black holes. For instance, his team has been able to predict the distinct signatures that should show up when black holes swallow stars. Just like a ringing bell, these black holes should set off vibrations in spacetime—known as “gravitational waves”—as they chow down. Khanna's simulations have characterized the frequency, duration, and other properties of these waves and this information will help observational efforts, such as those to be made by the [Laser Interferometer Gravitational-Wave Observatory \(LIGO\)](#), to find direct evidence of black holes.

But while Khanna's earlier simulations followed the evolution of black holes by using Einstein's general relativity as a framework, his latest work testing the predictions of loop quantum cosmology does not have this conventional underpinning. This raises a whole new set of challenges, says [Dan Christensen](#), at the University of Western Ontario, an expert on physics simulations and colleague of Khanna's. For instance, it is very tough to pinpoint what patch of spacetime you are looking at, because the nature and evolution of spacetime is itself being tested as part of the simulation, he explains.

Size also matters in these simulations. According to loop quantum cosmology, the pixels of spacetime are unthinkably tiny when compared with observable objects. “You might want to model a planet but your building blocks are 10^{-33} cm!” says Christensen.

Despite these difficulties, preliminary simulations have already shown that some drastic changes were needed in the loop quantum cosmology because the modeled universe did not match the features we see in the real universe around us. “The first equations we tested didn't work, especially for cosmology and black holes,” said Martin Bojowald, at Pennsylvania State University, one of the theoretical physicists behind the loop quantum cosmology. “Something was wrong and we saw that the system was unstable, which wasn't obvious when we derived the equations from a mathematical perspective.”

Bojowald and his colleagues hope that their revised equations will fare better in Khanna's next round of tests. If so, their refined theory could provide the blueprints for simulating conditions before our universe existed. But if not, loop quantum cosmologists will just have to keep trying to move farther up the PS3 leaderboard.



THE GRAVITY GRID

The PS3 network is the equivalent of 200 normal desktop computers.

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Holy Trinity equation.

$1/3 \text{ APPLE} + 1/3 \text{ oRANGE} + 1/3 \text{ oRANGE} = 1 \text{ APPLE/ORANGE}$

$1 + 1 = 2$

$1 \text{ odd} + 1 \text{ Even} = 2 \text{ ODD.}$

$2 + 2 = 4$

$2 \text{ ODD} + 2 \text{ EVEN} = 4 \text{ EVEN}$

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